

Matthew Cranny

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Systems engineer developing automated hardware tests, custom bench electronics, and embedded C firmware. Experience spans board qualification, instrumentation, international firmware deployment, and aerospace hardware assembly.

EXPERIENCE

PDF Solutions

March 2026 - Present

Systems Engineer

San Jose, CA

- Own qualification, calibration, and production acceptance across six board families using Python, PyQt6, bench instruments, SCPI, GPIB, and NI-DAQmx for electrical, analog, thermal, telemetry, and firmware testing.
- Transitioned board testing to a custom OpenTAP front end and OpenHTF back end, standardizing test workflows and interfaces to improve reliability and scalability.
- Developed a custom selector for four electrostatic correction boards; wrote C firmware with SCPI-style commands over RS-485 to route measurement outputs and Ethernet control for the active board.
- Investigated power-up IP assignment and connectivity failures surfaced by faster automated testing; used AMD Vivado and register-level analysis to diagnose hardware and firmware faults.
- Coordinated overseas and field teams for firmware deployments, arranging scan comparisons before and after upgrades to verify equal or better performance, supplying updated hardware, and guiding remote installation.
- Built a Flask and SQLite system of record covering hardware and firmware revisions for more than 90 boards; track configuration conflicts, deployment requirements, and upgrade progress.
- Characterized SEM settling with Python signal analysis and tuned inertia compensation, reducing time for oscillations to fall below 2 nanometers from 300 to 20 nanoseconds.

Amazon

May 2025 - July 2025

Area Manager Intern - ICQA

Turlock, CA

- Built Python API and analysis pipelines for Inventory Control and Quality Assurance; migrated anomaly detection to AWS Lambda, EventBridge, and Secrets Manager, reducing estimated annual vendor transaction risk by \$100,000.
- Automated labor reporting across fulfillment centers with VBA, eliminating over 120 hours of manual work each week.

Space Sciences Laboratory, UC Berkeley

June 2024 - May 2025

Aerospace Engineering Intern

Berkeley, CA

- Modeled structural components in SolidWorks for LuSEE-Night, a NASA-funded lunar radio observatory; fabricated and assembled satellite hardware in cleanrooms under contamination-control protocols.
- Assembled high-voltage step-up converters and installed flight-hardware wiring harnesses and passive components.

SELECTED ENGINEERING PROJECTS

Programmable Protocol Emulator ASIC

Independent Project

- Built a programmable Verilog timing engine with SPI host control, UART transmit firmware, and Python tooling for reproducible protocol debugging on IHP CMOS5L.
- Validated with 200 Python tests, six RTL simulation suites, and formal safety checks; implemented a routed layout targeting 25 MHz, with electrical closure still in progress.

Combat Robotics at Berkeley

August 2023 - May 2025

Technical Lead

- Built weapon and drive power systems integrating motor controllers, brushless motors, wiring harnesses, and battery management; designed packaging to protect electronics from shock and vibration.

EDUCATION

University of California, Berkeley

May 2026

Bachelor of Arts in Physics

Coursework: Instrumentation Laboratory, Advanced Experimental Laboratory, Electromagnetism and Optics, Quantum Mechanics, Statistical and Thermal Physics

TECHNICAL SKILLS

Hardware and Test: Oscilloscopes, custom test hardware, PyVISA, AMD Vivado, SolidWorks, cleanroom assembly

Software: Python, C, Verilog, SQL, LabVIEW, pytest, cocotb, Icarus Verilog, GitHub Actions, Linux